Batterjee Medical College
For Science and Technology

The pituitary gland Disorders

Internal Medicine Department

Learning Objectives

Knowledge

- Relate pathophysiological basics and etiology of pituitary gland disorders
- Describe epidemiology, manifestations, complications and management of pituitary gland disorders

Skills

- Interpret clinical manifestations and investigations of pituitary gland disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for pituitary gland disorders
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

Main Subtopics

- Hypopituitarism in childhood (causes, pituitary dwarfism, Frohlich's syndrome)
- Hypopituitarism in adults (panhypopituitarism, isolated hormone deficiencies, clinical features)
- Investigations and pituitary reserve tests (TSH, LH/FSH reserve tests)
- Treatment of hypopituitarism (surgery, radiotherapy, hormone replacement)
- Pituitary apoplexy (causes, clinical picture)
- Short stature (differential diagnosis, diagnostic approach, investigations, treatment)

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Hypopituitarism

Causes

1) Infarction of pituitary:

A. postpartum necrosis (sheehan syndrome)→ the most common cause: the anterior pituitary is the most susceptible to infarction in the post-partum period (P.P.Hge) because :

→ it is enlarged during pregnancy without increase in blood supply.

→ its blood supply is mainly venous.

B. Vascular diseases→ e.g. vacuities, DM.

2) Infection & granuloma: T.B. , syphilis, meningitis, sarcoidosis.

3) Idiopathic(congenital): deficiency of one or all ant. pituitary hormones.

4) Neoplastic: adenoma,craniopharyngioma,leukaemia, lymphoma, metastasis.

5) Physical agents : radiation, surgery, head trauma.

6) Miscellaneous :e.g.hemochromatosis

Clinical Picture

A. In childhood :

1) Pituitary dwarfism (Levi-Lorain syndrome)

- Proportionate dwarfism.
- Childish facies.
- Later on, hypogonadism may appear (with cryptorchidism in male).

N.B: There is ↓GH 'commonly idiopathic '. Rarely, inactive GH secretion

2) Frohlich 's syndrome (dystrophia adiposogenetalis)

- Dwarfism, hypogonadism.
- Obesity (in trunk, face sparing the extremities 'samboxa shaped obesity).
- Genu valgum.
- Hypothalamic disturbances as DI, polyphagia, hypersomnia.



Hypopituitarism with gonadotrophin failure in a 39-year-old man.

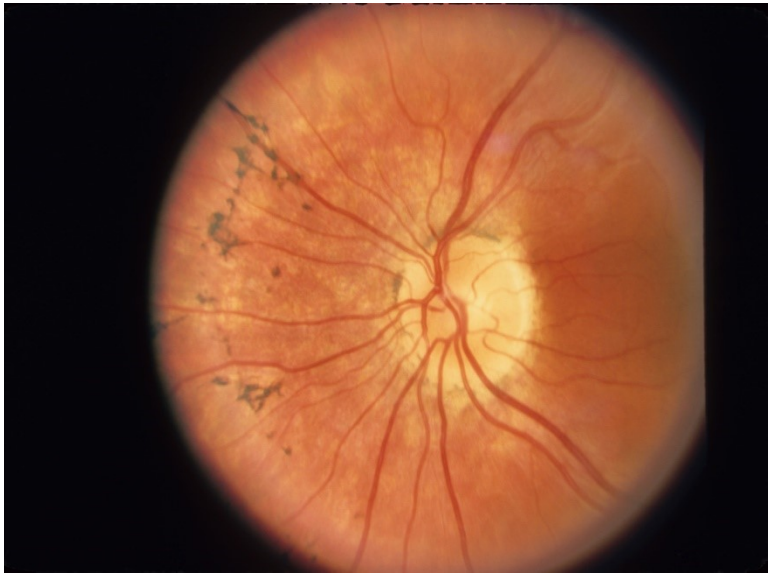
Note the extreme atrophy of the genitalia, the absence of body hair and the apparent gynaecomastia associated with obesity.



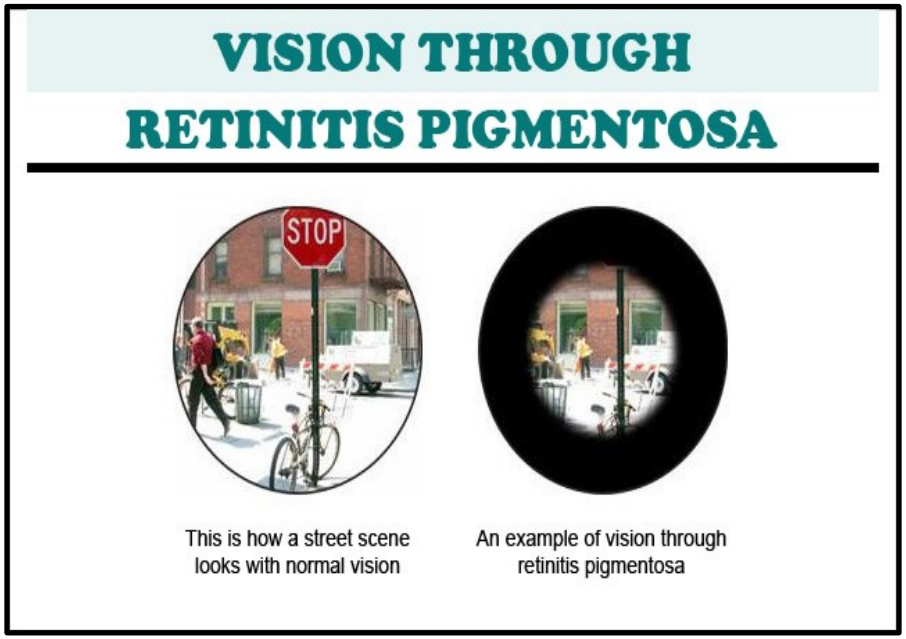
3) **Laurance-Moon-Biedle syndrome** :

Similar to Frohlich's syndrome +

- Mental retardation. & skull deformities.
- Retinitis pigmentosa.
- polydactyly, syndactyly .







B. In adults :

- Panhypopituitarism (Simmond's disease).
- Isolated hormone deficiency.

Clinical features :

A. Prolactin ↓→ failure of lactation.

B. Growth hormone ↓ → silent, rarely leads to hypoglycaemia.

C. Gonadotrophic hormones ↓ → ***Post-pubertal***

o Skin → fine wrinkled, loss of 2ry sexual hair 'progeria

o Sexual manifestations

- ↓ libido, impotence, testicular atrophy, oligospermeia
- Vaginal dryness, dyspareunia, 2ry amenorrhea & infertility +breast atrophy

D. TSH ↓→ 2ry hypothyroidism

> **Student pearl**
>
> Hypothalamic dopamine inhibits prolactin secretion, whereas TRH stimulates it.

E. ACTH ↓→ usually not marked & may cause hypoglycaemia, inability to withstand physical & psychic stress & lowering of resistance to infection

F. Pressure neurologic manifestations: →in case of pituitary tumour.

G. Coma : due to

- Hypothermia
- Hypoglycaemia or
- Pressure on brain stem or hypothalamus.

	2ry hypothyroidism	1ry hypothyroidism
severity	Less sever	More sever
skin	Fine wrinkled	More oedemotous
hair	Dry, less coarse	Dry, brittle, coarse
tongue	Not enlarged	enlarged
Visual field defects	May be present	abscent
Heart & blood pressure	Small heart & ↓ blood pressure & atrophy of viscera(splanchnomicria)	Enlarged heart & hypertension is common
Menstrual changes	amenorrhea	Menorrhagia(mostly)
Blood glucose	Usually low	Usually normal
Serum cholesterol	Normal	↑
TSH	↓	↑
Other anterior pituitary hormones	↓	normaol

	Addison's disease	Simmond's disease
pigmentation	present	Abscent (lack of lipotropin) + pallor (ACTH normally ++ erythropiosis) + depigmentation of areola in females
hypotension	Marked (+hypovolemia & hyperkalemia)	Less marked(due to continued release of aldosterone → no ↑ serum K or hypovolaemia)
Bowel habits	diarrhea	Constipation
ACTH	↑	↓ (+ other pituitary hormone ↓)

Investigations:

In addition to radiological findings of pituitary tumour, the following lab. Findings are present:

1.Tests for GH reserve :

Measurement of GH level during sleep & after exercise → the level should rise " > 6 ng / ml ", if not proceed to the following tests :

1- Insulin induced hypoglycaemia	2- L-dopa	3- Clonidine test
(0.1 U/kg I. V.)	(250 mg orally)	(50 ug for each l0 years of age above 5 years with a maximum of 25 0 ug orally)

Samples are taken fasting (before giving the drugs), 60, 90 minutes later → values > 6 ng / ml. exclude GH.

2. Tests for ACTH reserve :

- Insulin induced hypoglycemia : → ↑ in cortisol level (> 10 ug/dl) above the basal value indicates normal response.

3.Tests for TSH reserve :

- TRH test → 500 ug IV of TRH cause doubling of the basal TSH level. Low response is obtained in 2ry hypothyroidism (& in thyrotoxicosis).

4.Tests for LH, FSH reserve :

A. Normal gonadal function, i.e. normal cyclic menses or normal sperm count exclude hypothalamic - pituitary disease.

B. Basal LH & FSH.

C. LHRH [GnRH] test (l00 ug IV.)

D. Clomiphene citrate test (oestrogen antagonist)→ The normal response in both (c & d) is doubling the baseline LH & about 50 % ↑ of FSH.

Treatment

A. *Surgery* → if there is space occupying lesion (through transfrontal or trans- sphenoidal approach).

B. *Radio-therapy :* →indicated in :

- Incomplete removal of the tumour surgically.
- Patients not candidates for surgery.

C. *Hormone replacement* :

1. Glucocorticeid : e.g. prednisone 5 mg in the morning & 2,5 mg at noon, orally.
2. L-thyroxine: → 100- 250 ug / day.

   It is given after cortisol replacement is begun to avoid precipitation of adrenal failure.
3. GH. → 3 injections / week (before the age of epiphyseal closure)

4.Gonadotrophins

- if fertility isn't needed
  - o In male → Long acting testosterone LM.
  - o In female→ Cyclic estrogen & progesterone
- If fertility is needed (in either sex):
  - o Human menopausal gonadotrophin (HMG)→ Having FSH activity
  - o Human chorionic gonadotrophins (HCG)→ Having LH activity.

→For several months.

5. ADH → desmopressin by nasal insufflation or pitressin tannate IM

Clinical pearl

Patients with pituitary incidentalomas should be evaluated for hypopituitarism and hormonal hypersecretion syndromes. [8]

Caution

About a third of patients with pituitary adenomas have associated hypopituitarism; consider screening for hormone deficiencies in all patients with a pituitary mass. [9]

Caution

Pituitary adenomas may bleed spontaneously, causing pituitary apoplexy (i.e., pituitary tumor apoplexy); this manifests with severe headaches, visual symptoms, cardiovascular collapse, and/or acute secondary adrenal insufficiency. [10]

PITUITARY APOPLEXY

Causes:

1. Hemorrhage in pituitary tumour
2. Cerebro-vascular accident (particularly in diabetics).
3. Sickle cell anemia.

> **Clinical pearl**
>
> Pituitary apoplexy, which results in acute hypocortisolism and hypothyroidism, can manifest with severe headache, sudden hypotension, and hypovolemic shock.

Clinical picture:

1. The most common initial symptom is a sudden headache, often associated with a rapidly worsening visual field defects or double vision caused by compression of nerves surrounding the gland.
2. This is followed in many cases by acute symptoms caused by lack of secretion of essential anterior pituitary hormones, predominantly adrenal insufficiency. Rarely apoplexy produces autohypophysectomy with cure of acromegaly or Cushing disease.
3. Manifestations of ↑ ICT.
4. Manifestations of meningeal irritation.
5. Confusion that may progress to coma 'with hyperpyrexia' & if not urgently treated end fatally.

Investigations :

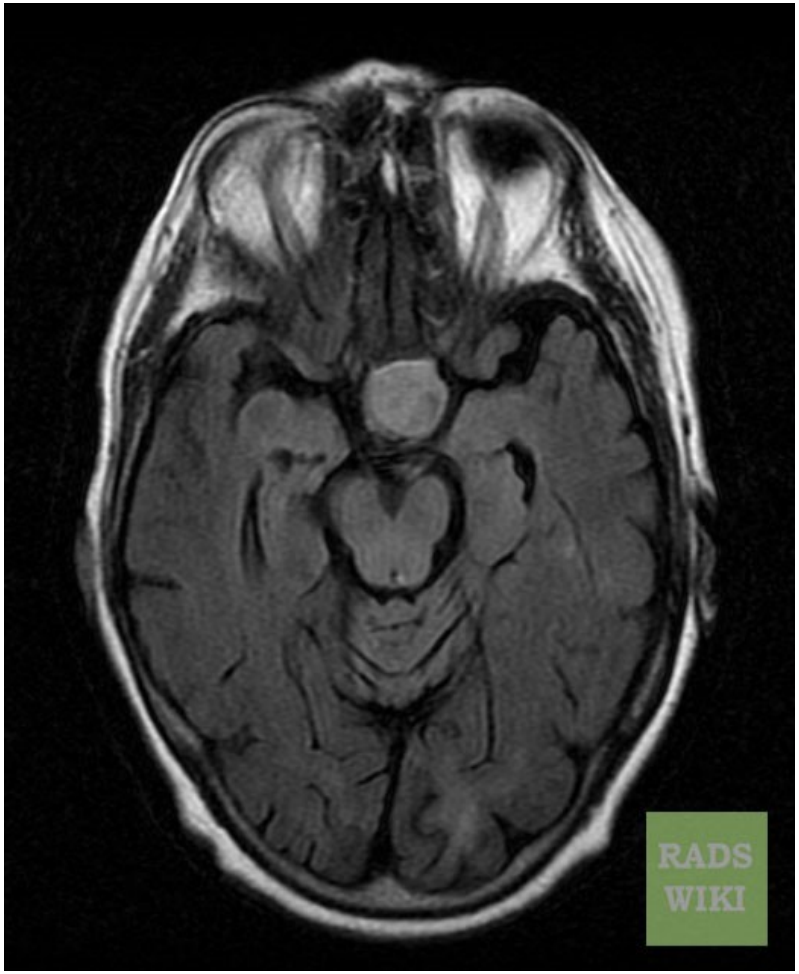
1.CT. scan → is diagnostic.

2.C.S.F. → bloody.

Treatment:

1. Glucocorticoids e. g. dexamethasone.

1. Surgical decompression may be needed.



SHORT STATRUE/ Causes

Primary defects : i.e.“ defects intrinsic to the skeletal system”

1.Skeletal dysplasia (condrodystrophies) e.g. achondroplasia

2.Chromosomal abnormalities:

- o Down syndrome. (trisomy 21)
- o Turner syndrome.

3.Inborn error of metabolism :

- o Glycogen storage disease.
- o Mucopolysaccaridosis.

4.Genetic short stature: e.g. pygmies.

Secondary defects i.e. “ defects outside the skeletal system ”

1.Intrauterine growth retardation :

- o Infection.
- o Placental insufficiency.
- o Maternal illness.

2.Idiopathic (constitutional) growth delay “ most common cause ”

3.Malnutrition → e. g. Kwashiorkor, marasmus, etc.

4.GIT diseases :

- o Celiac disease.
- o Chronic liver disease.
- o Cystic fibrosis.
- o IBD

5.Renal diseases: e.g. chronic parenchymal diseases, RTA.

6.Cardio-respiratory diseases → e. g. congenital heart diseases, bronchial asthma.

7.Endocrine diseases:

- o Hypopituitarism & isolated GH or GHRH ↓, (or ↓peripheral tissue response to GH).
- o Pseudohypoparathyroidism.
- o Precocious puberty (Androgen / estrogen excess).
- o Cushing syndrome.
- o Cretinism & Juvenile myxedema (4- 12 years).

8.Psychological deprivation.

Diagnosis

Height within 2 SD of the mean of normal for age may be normal. But if < 3 SD → a thorough evaluation of short stature should be done.

1.Body proportions : Short limbs compared to the trunk suggest long standing hypothyroidism or achondroplasia.

2.Height -to - weight ratio

3.Specific physical findings :

a. Achondroplasia : macrocephaly, short limbs in comparison to the trunk.

b. Cretinism: → dry skin, coarse hair, depressed nasal bridge, big lips, large protruding tongue, etc.

c. Cushing syndrome: → central obesity, striae, HPN,..

d. Turner syndrome : → webbed neck, shield chest, coarctation of aorta, delayed sexual development.

e. GH ↓: (see before).

f. Pseudohypoparathyroidism :

- Moon face - Obesity - Short metacarpals - Mental retardation.

Investigations

1. Hormonal assay: e. g. thyroxin, cortisol, GH. (and response of GH to exercise, clonidine test → see before).
2. Renal assessment: urea & creatinine.
3. Investigations for malabsorption, C.V.S.
4. Chromosomal karyotype (in gonadal dysgenesis).
5. Estimation of:
    - Ca (↓ in pseudohypoporathyroidism).
    - P (↓ in vitamin D resistant rickets).
    - HCO3 (↓ in RTA)
6. Skull x-ray :→ to detect CNS lesions (e. g. cranio-pharyngiomo).
7. Determination of bone age (hands): delayed in 2ry defects (but not retarted in 1ry defects).

Treatment

1.Treatment of the cause : (as early as possible)

e.g. Cushing's syndrome, hypothyroidism,….

2.GH ↓: → synthetic GH (produced by recombinant DNA technology 'genetic Engineering') → 0.1 unit / kg S.C. 3 times / week.

NB: In lry defects, potential for bone growth is severely impaired unlike 2ry defects.

Case Scenario

A 39-year-old woman with a history of migraine headaches is brought to the emergency room because of a severe, sudden-onset, throbbing headache and double vision for 1 hour. She says that she has been having frequent headaches and has not had her period in several months. Her blood pressure is 93/61 mm Hg. Visual field testing shows a bilateral visual field defect in the lateral quadrants.

Which of the following is the most likely diagnosis?

a) Transient ischemic attack
b) Pituitary apoplexy
c) Sheehan syndrome
d) Migraine with aura
e) Subarachnoid hemorrhage
f) Cluster headache

Summary



References

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Thank You